

Project Brief:

Robotic Systems for Forest Management and Environmental Monitoring

About this Document

This brief introduces the overall project that you and your team will be working on throughout the semester. It outlines the project scenario, your role, the broad development phases, and the general expectations for your team.

The **teaching staff will act as the project's client and stakeholder**, evaluating your work from a stakeholder perspective at key stages of development. Further details, including technical requirements, submission formats, and marking criteria, will be provided progressively in **individual assessment briefs**. Your submissions across the semester will collectively form a Design Portfolio documenting your team's project development and your individual reflection on the process.

Project Vision

You are part of a small, specialised robotics company. Your team has recently been approached by a **client** working in **forest and environmental management** – this might be a park ranger, farmer, fire fighter, law enforcement, ecologist, conservationist, or land management organisation.

The client is interested in learning how **robotic systems** – including ground and/or aerial robots equipped with **sensing, autonomous navigation, and decision-making capabilities** – could help with the **monitoring, management, and tracking** of forest environments.

They have very little experience with robotics, but are curious and motivated. They've engaged your company to investigate and demonstrate the potential of such systems for applications that they are interested in. If the outcomes of the project are promising, they may seek to continue the collaboration.

Your role is to act as a professional robotics development team: **design, implement, test, and communicate** a functioning robotic system in simulation that addresses the client's needs – while showcasing your team's technical skills and project planning capabilities.



Figure 1: A futuristic vision of ground and aerial robots supporting forest management and environmental monitoring.

Team Structure

- Students will work in teams of **4 or 5** students
 - Teams of 5 are expected to make appropriate use of the additional team capacity, for example through stronger technical depth, more extensive evaluation, broader integration, or a more ambitious but still feasible scope.
- Each team functions as an independent **robotics company**
- You will divide responsibilities across system components, taking on both **lead and support roles**
- **Collaboration, accountability, and communication** will be essential throughout the project
- You are welcome and encouraged to engage with other teams in the subject, but the ideas, design decisions, implementation, and evaluation presented by your team **must be your own**, and your team must be able to **clearly communicate and justify your solution** to the stakeholder.

Project Development Phases

The project is structured around three broad development phases. These phases are not strict end points; your design, implementation, testing, documentation, and communication should continue to evolve throughout the semester.

1. Problem Framing and Design Concept (Weeks 1-4)

“We want to understand how robots might be used in forest environments – both in the short term and long term. What kind of problems could they help solve?”

You will:

- Brainstorm possible use cases, such as trail inspection, fire detection, mapping, biodiversity surveys, environmental monitoring, or other forest management applications.
- Consider both achievable short-term tasks and more ambitious long-term possibilities.
- Reference existing systems or research (e.g. conservation drones, autonomous mapping platforms)
- Identify specific technical and environmental challenges in forest settings

By the end of this phase:

- Your team must **select one key application/scenario** as the focus of your project moving forward
- This selection should be **justified** based on:
 - Its **value or importance** to real-world stakeholders
 - Its **feasibility** to achieve and demonstrate within the 12-week project timeline, using the capabilities of your team
- Be realistic and ambitious: choose something achievable within the subject timeline that still pushes your team’s technical skills.
- Your proposed direction must be **discussed and negotiated** with the client/stakeholder (i.e. teaching staff) before proceeding.

2. Working Prototype and System Development (Weeks 5-9)

“We’d like to see a robot system working in simulation. What does it take to make that happen?”

Teams are expected to begin technical implementation early in the semester and to build evidence of progress through working prototypes, experiments, testing, and regular integration.

You will:

- Build a working simulation of a robot system operating in a representative forest-like environment

- The teaching team will provide a starting simulation environment, including ground and aerial robotic platforms, sensors, and example forest-like worlds. You are expected to use this as a starting point, then adapt, extend, or replace parts of the system as needed for your chosen project.
- Demonstrate essential capabilities across the three core technical components:
 - **Perception:** sensing, detecting, mapping, estimating, classifying, or otherwise interpreting relevant information from the environment.
 - **Decision making:** selecting actions, planning motion, allocating tasks, responding to environmental information, or making mission-level choices.
 - **User interface:** presenting useful information to a user or operator, such as maps, alerts, status information, mission progress, or other decision-support outputs.
- Develop a **clear system architecture**, outlining:
 - Key components, including perception, decision making, user interface, and any supporting elements such as control, data processing, communication, simulation, or evaluation tools.
 - Team roles and responsibilities (who leads, who supports)
 - Integration plan and internal testing milestones
- Use a combination of:
 - **Off-the-shelf components:** standard robots, open-source libraries, baseline planners
 - **Custom components:** software, system behaviours, evaluation methods, simulation features, or user-interface elements developed by your team that go beyond the provided starting point. Creating meaningful custom components is strongly encouraged because it demonstrates your technical capability and gives the client a clearer reason to continue the collaboration.
- **Update and extend your proposal** to outline what you aim to achieve in the following phase

3. Advanced Functionality and Final Demonstration (Weeks 10-12 and exam period)

“Once the basics are working, what more advanced tasks could robots do for us?”

This phase should build on the working prototype from Phase 2. The goal is not only to add more features, but to improve the reliability, integration, evaluation, and clarity of the system.

You will:

- Strengthen the system from phase 2 by **fixing any outstanding issues**
- Update your design to respond to any requests or “**surprises**” provided by the stakeholder
- Extend the system to showcase **more ambitious or intelligent behaviours**
- Think **creatively** about how robotic systems could be used in forest management – not just what they *can* do now, but what they *should* do in the future
- All teams are expected to continue strengthening the core technical components of perception, decision making, and user interface. Beyond these core components, teams are encouraged to pursue creative extensions that suit their chosen scenario and demonstrate technical ambition. Possible extension pathways include, but are not limited to:
 - Multi-robot coordination
 - Use of different mobility types, such as aerial robots
 - More complex or dynamic simulation environments
 - Adaptive exploration
 - Anomaly detection
 - Terrain analysis
 - 3D reconstruction
 - Segmentation of key environmental features
 - High-level mission planning

- Creative features that go beyond the standard expected technical components
- The specific **scenarios to be demonstrated** will be **negotiated with the stakeholder** based on each team's direction and progress
- **Communicate** and **demonstrate** your solution and findings to the project stakeholder

What Makes a Strong Project?

A strong project will not simply describe an idea; it will demonstrate a functioning robotic system and provide evidence that the system has been designed, implemented, tested, evaluated, and improved over time.

Successful teams will balance ambition with feasibility. A smaller system that works reliably, is well tested, and is clearly evaluated is usually stronger than a large concept that remains mostly theoretical.

The strongest projects will show technical depth, creativity, thoughtful system integration, meaningful evaluation, and clear communication of what the system can and cannot do.

Assessment & Deliverables

This project is worth **100% of your final subject mark** and will be assessed through a staged Design Portfolio, including **written documentation, presentations, technical demonstrations, individual reflection, and evidence of team accountability**.

Assessment tasks are staged across the session to support iterative progress. You will receive detailed assessment briefs with specific requirements, submission formats, and grading criteria for each component.

Proposing Other Themes

The default context for this project is forest management and environmental monitoring. In exceptional cases, teams may propose an alternative but closely related context, such as space robotics or another outdoor-based robotics application. Any alternative context must be approved by the Subject Coordinator, must be justified for a clear reason, and must involve similar overall goals and technical challenges.

Final Notes

- Treat this as a **real client project**: professionalism, clarity, and follow-through matter
- **Start technical development early**. Strong outcomes usually come from regular implementation, testing, integration, and refinement, not from a last-minute rush.
- Be **flexible**: projects evolve, problems emerge, plans change.
- **Leverage your team**: good outcomes come from strong collaboration, clear responsibilities, and shared accountability, not individual heroics
- **Take ownership of your work**: successful projects are those where teams make thoughtful technical choices, provide convincing evidence of progress, and communicate their solution clearly.
- **Be creative, but stay grounded**: ambitious ideas are encouraged when they are supported by realistic implementation and evaluation.

We're excited to work with you and see where your ideas take you!